Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

991
-2 -0.20%
Nov 29th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

1103
-2.02 -0.18%
Nov 29th, 2023

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

907
2 0.21%
Nov 29th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

135.78
0.70 0.52%
Nov 29th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

157.09
0.66 0.42%
Nov 29th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

1160
165 16.58%
Week Ending Nov 24th, 2023

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2401 (Jan) RMB/t (3pm close)

955.50
4.50 0.47%
Nov 29th, 2023 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
December 23 USD/dmt

128.55
1.25 0.98%
Nov 29th, 2023 (5.30 pm Print)

SHFE Rebar
RB2401 (Jan) RMB/t

3896
3 0.08%
Nov 29th, 2023 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

26.96
-0.02 -0.06%
Nov 28th, 2023

C5, W. Australia - Qingdao USD/t

11.20
-0.58 -4.97%
Nov 28th, 2023

Steel Price

Steel Rebar (China Domestic) RMB/t

4060
-20 -0.49%
Week Ending Nov 24th, 2023

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

110.49
0.64 0.58%
Week Ending Nov 24th, 2023

Steel Inventory in China
million tonnes

9.60
-0.20 -2.09%
Week Ending Nov 24th, 2023

Steel Price

Steel HRC (China Domestic) RMB/t

3960
0 0.00%
Week Ending Nov 24th, 2023

IRON ORE PORT STOCK INDEX (IOPI)

Nov 29th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	991	-2.0	-0.20%	855	879	756	1003	130.45	-0.09	-0.1%	113.11	119.42	97.78	131.57
IOPI58	58% Fe Fines	907	1.9	0.21%	758	762	650	907	120.33	0.43	0.4%	101.01	103.87	84.02	120.14
IOPI65	65% Fe Fines	1103	-2.0	-0.18%	972	1013	881	1114	145.58	-0.07	0.0%	129.13	138.38	115.00	146.65

IRON ORE SEABORNE INDEX (IOSI)

Nov 29th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	135.78	0.70	0.52%	113.87	113.80	99.20	146.75
IOSI65	65% Fe Fines	157.09	0.66	0.42%	126.73	126.63	109.60	157.09

MARKET COMMENTARY

DCE iron ore futures fall today , the main contract I2401 closed 955.5, an decrease of 0.47% throughout the day; Traders ship according to market conditions. Inquiries from steel mills have decreased, resulting in poor purchasing enthusiasm. PBF at Shandong port dealt 975-979 yuan/mt; increased 5 yuan/mt over yesterday. PBF at Shandong port dealt 983 yuan/mt; decreased 5 yuan/mt over yesterday.
Yesterday, iron ore experienced a significant decline, and steel mills increased their restocking volume. Today's inquiries have significantly decreased, with only some traders who delivered at the end of the month closing deals. In addition, according to SMM tracking research, 3 blast furnaces were repaired and 2 blast furnaces resumed production in this cycle. And the maintenance of the blast furnace has a large volume, resulting in a significant decrease in iron production. The operating rate of blast furnaces according to SMM statistics is 90.63%, a decrease of 0.15% compared to last week. The utilization rate of blast furnace production capacity is 91.45%, a decrease of 0.51% compared to the previous period. The daily average iron production of the sample steel plant was 2.1881 million tons, a decrease of 12000 tons compared to the previous period. Combined with regulatory suppression, it is expected that mining prices will continue to fluctuate weakly in the short term.

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Nov 24th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	1160	165	16.6%	959	1024	842	1160	146.30	22.06	17.76%	122.20	134.92	105.17	146.00

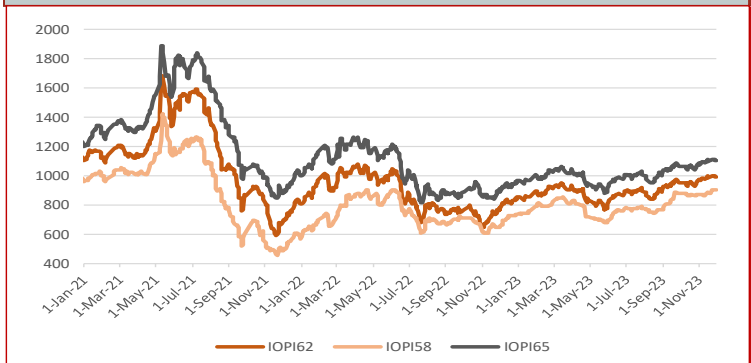
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Nov 24th, 2023				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	1155	2.1%	779	1645	161.93	2.12%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	1175	2.2%	780	1630	164.73	2.17%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	870	2.4%	620	1310	121.97	2.35%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	1224	1.7%	800	1752	171.60	1.66%	0.00	272.32

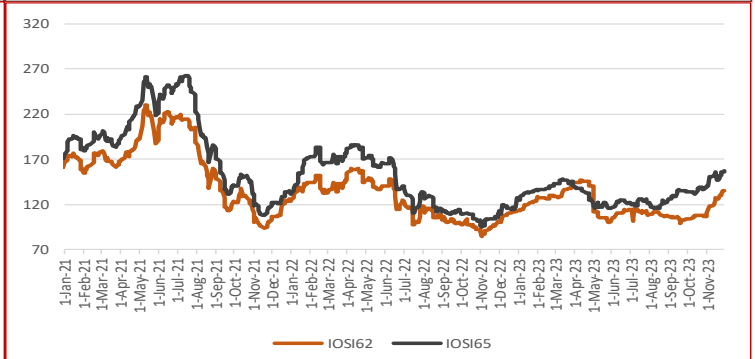
Week Ending Nov 24th, 2023	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	1058.13	-8.86%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =6.87528

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Nov 29th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	July	August	September	October	MTD	QTD	YTD	July	August	September	October	MTD	QTD	YTD
IOPI62	62% Fe Fines	896	873	945	947	855	849	879	116.58	113.28	122.79	123.14	113.11	112.73	119.42
IOPI58	58% Fe Fines	777	759	838	869	758	754	762	101.68	99.13	109.67	113.94	101.01	100.66	103.87
IOPI65	65% Fe Fines	1008	985	1057	1059	972	968	1013	131.57	128.31	137.74	138.13	129.13	129.03	138.38

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Nov 29th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	July	August	September	October	MTD	QTD	YTD
IOSI62	62% Fe Fines	108.57	109.02	104.25	106.44	112.14	112.92	113.80
IOSI65	65% Fe Fines	147.39	119.61	132.40	135.69	122.86	125.64	126.63

FREIGHT RATES

Nov 28th, 2023		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	11.20	-0.58	-4.97%	3.57	16.77
Tubarao - Qingdao	C3	26.96	-0.02	-0.06%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Nov 24th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	July	August	September	October	MTD	QTD	YTD	July	August	September	October	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	980	951	1043	1113	933	955	1017	122.51	118.96	130.25	139.38	119.96	122.06	134.98

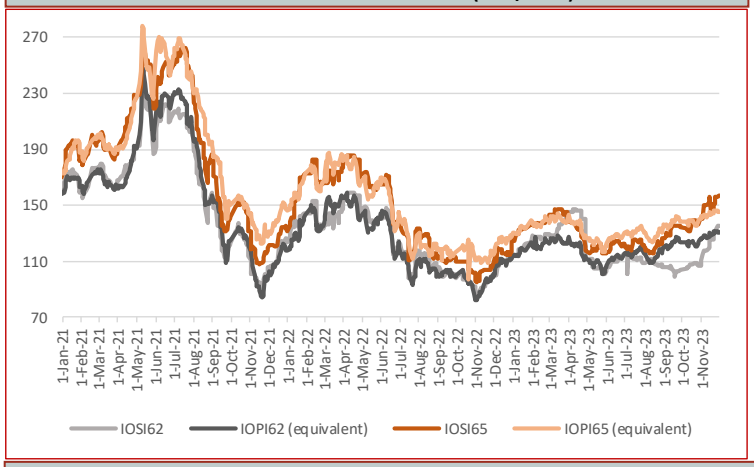
IRON ORE INDEX PREMIUMS/DISCOUNTS

Nov 29th, 2023		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-84	-8.52%
IOPI65	65% Fe Fines	112	11.27%

Nov 29th, 2023		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	21.31	15.69%

COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Nov 29th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	934	-2	-57
SIMEC Fines	866	-2	-125
PB Fines	957	-2	-34
Newman Fines	961	-2	-30
MAC Fines	941	-2	-50
Jimblebar Blended Fines	857	-2	-134
Carajas Fines	1172	-2	181
Brazilian SSF	958	-2	-33
Brazilian Blend Fines	971	-2	-20
RTX Fines	875	-2	-116
West Pilbara Fines	902	-2	-89

Nov 29th, 2023	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	131.25	0.70	-4.53
SIMEC Fines	127.75	0.70	-8.03
PB Fines	132.00	0.70	-3.78
Newman Fines	134.85	0.65	-0.93
MAC Fines	132.00	0.70	-3.78
Jimblebar Blended Fines	124.35	0.70	-11.43
Carajas Fines	165.30	0.70	29.52
Brazilian SSF	139.50	0.70	3.72
Brazilian Blend Fines	141.15	0.70	5.37
RTX Fines	125.65	0.70	-10.13
West Pilbara Fines	130.00	0.70	-5.78

Nov 29th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	907	0	0
FMG Blended Fines	918	0	11
Robe River	919	0	12
Western Fines	922	0	15
Atlas Fines	915	0	8
Yandi	899	0	-8

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	10.00	0.00
	High Grade Fe 63 - 64%	5.00	0.00
	High Grade Fe 64 - 65%	5.00	0.00
	High Grade Fe 65 - 65.5%	5.00	0.00
	Low Grade Fe	28.00	-1.00
1% Alumina	High Fe Grade Al <2.25%	19.00	0.00
	High Fe Grade Al 2.25-4%	16.00	0.00
	Low Fe Grade Al <2.25%	72.00	6.00
	Low Fe Grade Al 2.25-4%	37.00	-3.00
1% Silica	High Fe Grade Si <4%	5.00	0.00
	High Fe Grade Si 4-6.5%	20.00	5.00
	Low Fe Grade	25.00	1.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	5.00	0.00
	High Fe Grade 0.115%<P<0.15%	5.00	0.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

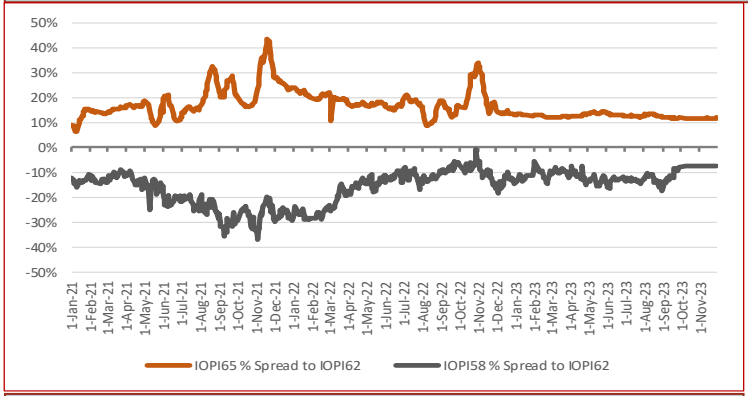
	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	3.50	0.00
	High Grade Fe 63 - 64%	2.75	0.00
	High Grade Fe 64 - 65%	2.75	0.00
	High Grade Fe 65 - 65.5%	2.75	0.00
1% Alumina	High Fe Grade Al <2.25%	8.00	0.00
	High Fe Grade Al 2.25-4%	0.75	0.00
1% Silica	High Fe Grade Si <4%	1.00	0.00
	High Fe Grade Si 4 - 6.5%	5.00	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	0.25	0.00
	High Fe Grade 0.115%<P<0.15%	0.75	0.00

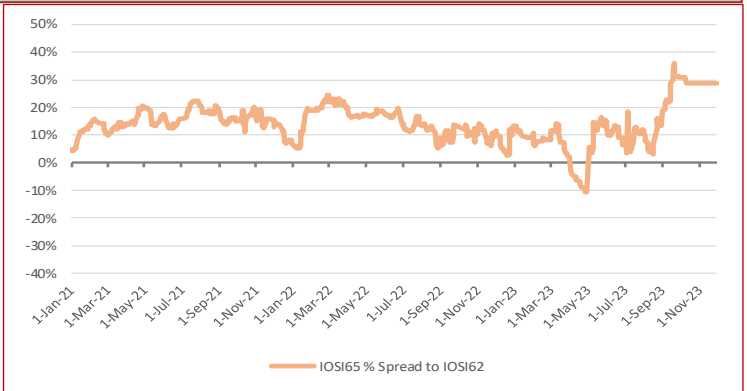
Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-20.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-10.00	0.00	Jingtang	-10.00	0.00	Majishan	0.00	0.00	Taicang	-30.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-30.00	0.00

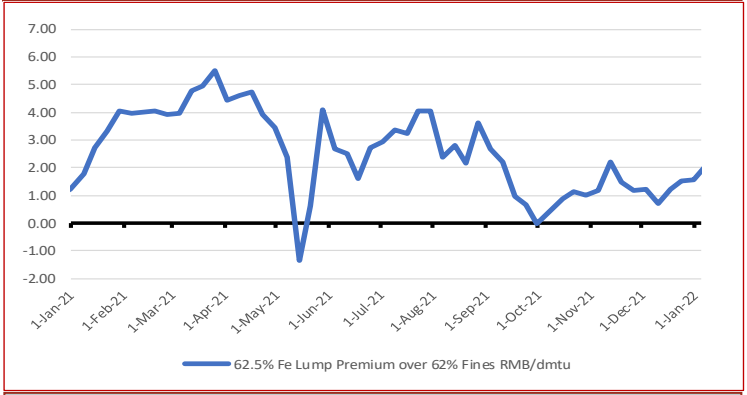
COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

IRON ORE INDEX PREMIUMS/DISCOUNTS

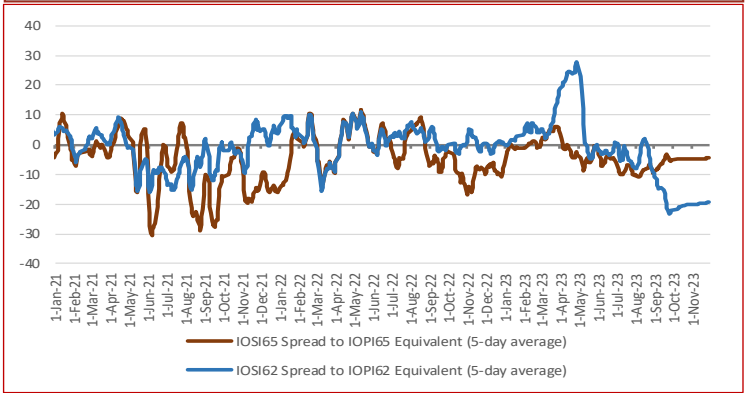




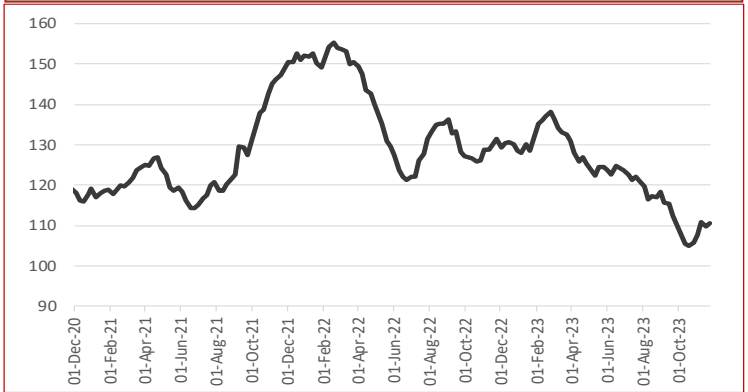
WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)



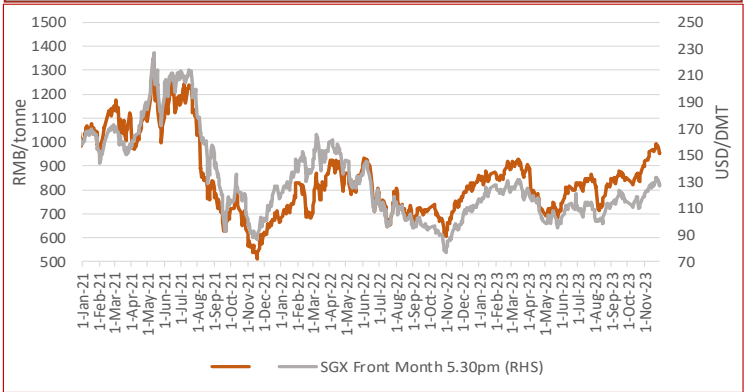
IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)



FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

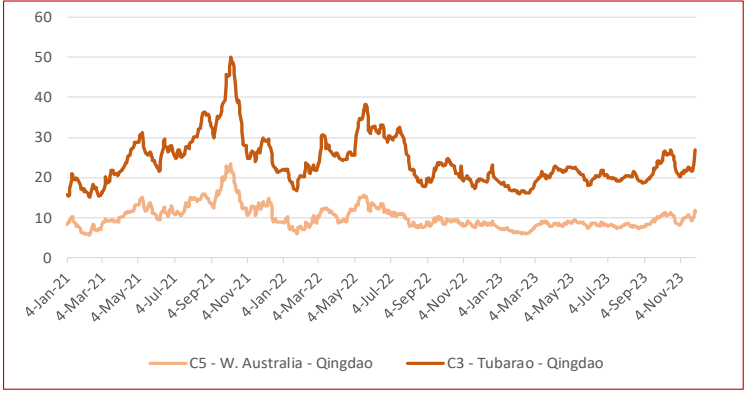
Week Ending Nov 24th, 2023

Province	This week	Change %	Low [2]	High [2]
Jingtang	8.88	3.02%	8.41	19.20
Qingdao	23.27	16.76%	9.41	26.24
Caofeidian	8.55	-6.46%	7.56	16.29
Tianjin	7.66	16.24%	6.49	12.97
Rizhao	12.94	5.89%	9.44	19.26
Total (35 Ports)	110.49	0.58%	98.80	155.39

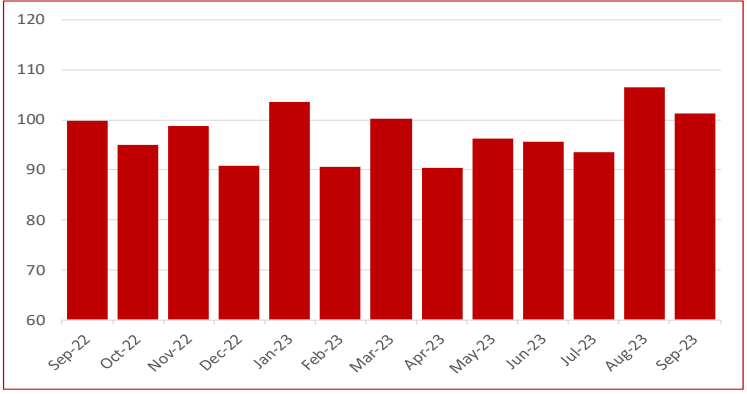
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Nov 29th, 3pm close			Nov 29th, 5:30pm		
Contract	I2401	Change	Change %	Dec. 23	Change	Change %
Closing Price	955.50	4.50	0.47%	128.55	1.25	0.98%
Vol traded ('000 lots)	21.09	-14.62	-40.94%	6.70	-1.18	-15.02%
Open positions ('000 lots)	57.67	-3.53	-5.77%	33.56	-0.73	-2.13%
Day Low	949.0	8.00	0.85%	127.50	0.20	0.16%
Day High	964.5	-15.50	-1.58%	130.60	-2.40	-1.80%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2023/11/24	Change	Change %
ReBar HRB400 ϕ18mm	4,060	-20	-0.49%
Wirerod Q300 ϕ6.5mm	4,290	-10	-0.23%
HRC Q235/SS400 5.5mm*1500*C	3,960	0	0.00%
CRC SPCC/ST12 1.0mm*1250*2500	4,810	20	0.42%
Medium & Heavy Plate Q235B 20mm	3,980	0	0.00%
GI ST02Z 1.0mm*1000*C	5,060	0	0.00%
Colour Coated Plate	7,000	50	0.72%

CHINESE STEEL MILL PROFITABILITY

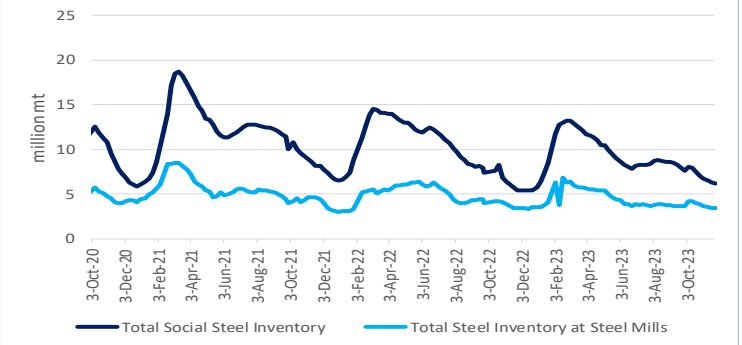
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	118.10	0	Mmi CFR Equivalent index for 1st Feb
Coke	2,560	110	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	2,910	0	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,319	49	Q234, incl. tax
Rebar cost - Blast furnace	3,977	55	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	-7	-65	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	4,038	60	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	-58	-40	based on Shanghai prices, incl. tax

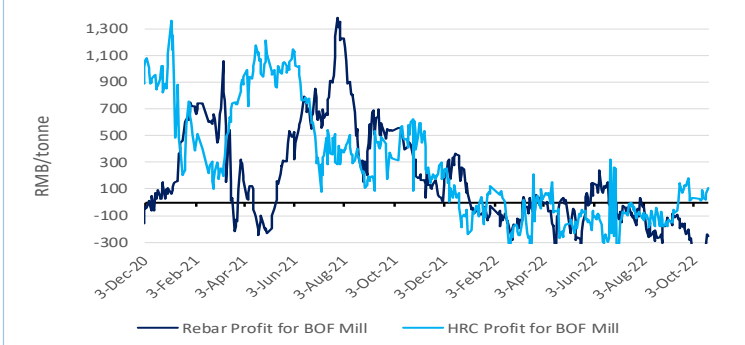
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

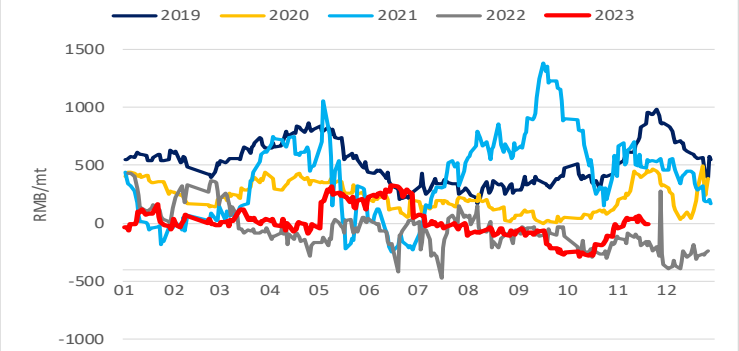
CHINESE STEEL INVENTORIES



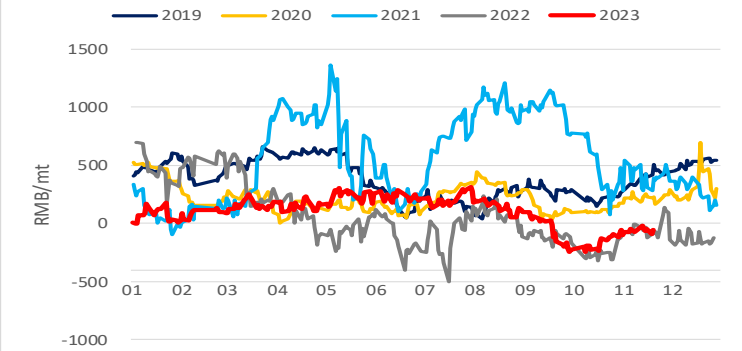
CHINESE STEEL MILL PROFITABILITY



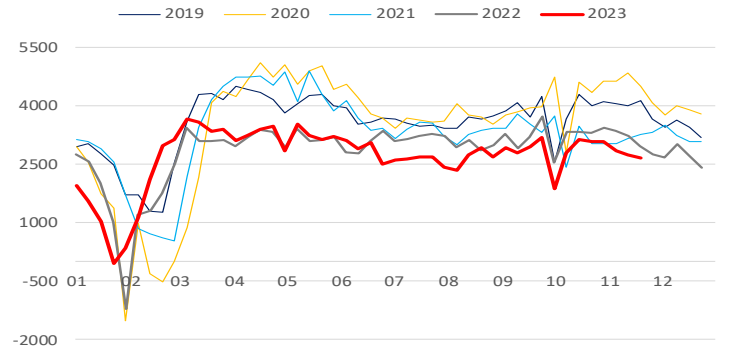
CHINESE STEEL MILL PROFITABILITY—Rebar



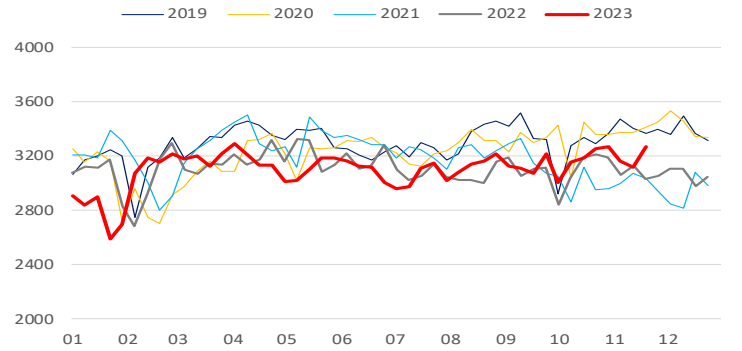
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



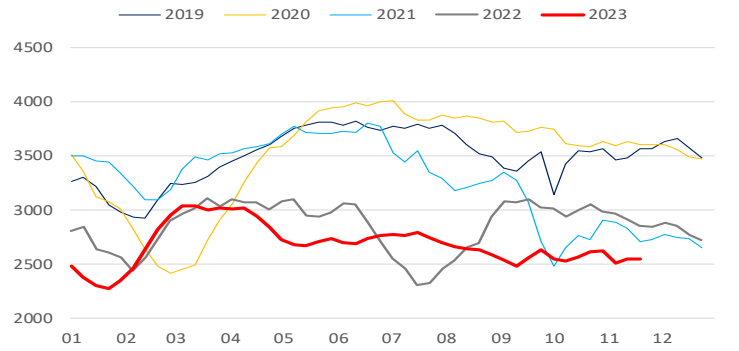
CHINESE STEEL CONSUMPTION—Rebar



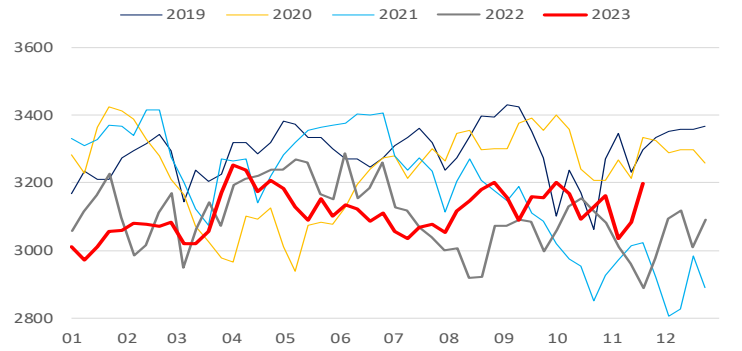
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Nov 29th, 2023	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Nov 29th, 2023	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

CONTACT US

MMI Singapore Office Details: Level 28, Manulife Tower, 8 Cross Street Singapore. Tel: + 65 6850 7629 E: prices@mmiprices.com
SMM Singapore Office Details: Level 28, Manulife Tower, 8 Cross Street Singapore. Tel: + 65 6850 7630 E: service.en@smm.cn
SMM Shanghai Office Details: 9th FL, Building 9, Lujiazui Software Park, No.20, Lane 91, Pudong, Shanghai Tel: +86 021 5155 0306 E: service.en@smm.cn

Visit www.mmiprices.com for full index price histories and archive of daily reports

This information has been prepared by Metals Market Index ("MMi"). Use of the information presented here is at your sole risk, and any content, material and/or data presented or otherwise obtained through your use of the information in this document is at your own discretion and risk and you will be solely responsible for any damage to you personally or your company or organisation or business associates whatsoever which in anyway results from the use, reliance or application of such content material and/or information. Certain data has been obtained from various sources and any copyright existing in such data shall remain the property of the source. Except for the foregoing, MMi retains all copyright within this document. The copying or redistribution of any part of this document without the express written authority of MMi is forbidden.

COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED